Space Debris Collision Avoidance — Team Execution Plan

Team size: 5 (3 AI, 1 Web, 1 Speaker)
Pre-SIH runway: 5 days
SIH duration: 24 hours (live)

This plan assumes the core physics/ML engine is built and validated *before* SIH starts, so the 24-hour window goes toward integration, live-data hardening, polish, and demo readiness — not toward writing SGP4 propagation from scratch under time pressure.

One thing worth checking before you commit to this approach: some SIH tracks explicitly require the working solution to be built during the event window, and judges sometimes ask teams to show commit history or a live build log. If your specific SIH track has that rule, treat the pre-SIH work as a **prototype/research spike** you rebuild-and-extend live, rather than code you ship as-is — the architecture and validated math are still yours to reuse either way, but check the rulebook so this doesn't become a disqualification risk.

---

1. Roles and Ownership

Person	Primary Specialty	Owns	Backup Role
AI-1	Orbital Mechanics	TLE ingestion, SGP4 propagation, coordinate frame transforms (TEME→ECI→ECEF)	Can help AI-2 on screening logic
AI-2	Conjunction & Probability Math	Coarse+fine conjunction screening, Pc calculation (Foster's 2D method + Monte Carlo fallback)	Can help AI-3 on feature engineering
AI-3	ML & Decision Support	Feature engineering, LightGBM risk ranking model, SHAP explainability, TabPFN comparison, maneuver advisory (delta-v optimization)	Can help Web on API contracts
Web	Frontend + Backend Integration	FastAPI backend, Cesium.js/React globe, dashboard, deployment (Docker Compose)	Can help any AI role debug data pipes

Person	Primary Specialty	Owns	Backup Role
Speaker	Narrative + Delivery	Pitch deck, demo script, problem framing, judge Q&A prep, backup demo video	Can do QA/testing, data labeling, docs, timekeeping during SIH

Everyone can do everyone else's work in a pinch — this table is about default ownership and who makes the final call on their layer, not a hard wall. The Speaker role is explicitly the lightest-loaded during pre-SIH days (nothing to present yet), so treat Speaker as a floating QA/documentation/support resource for the AI trio in days 1–3, then have them ramp up narrative work from day 3 onward once there's something real to narrate.

Why this split works for a 5-person team (not the blueprint's 4)

The original blueprint assumed 4 roles (Orbital, Conjunction/Math, ML, Frontend/Viz) with backend split across all four. With 5 people, backend ownership moves fully onto Web (freeing the AI trio to stay in their lane), and you gain a dedicated Speaker who can also double as your integration tester — someone who isn't heads-down in their own module is often the first to notice when two modules don't actually agree on a data format.

---

2. Interface Contracts (lock this on Day 1)

Before anyone writes pipeline code, agree on the data shape crossing each boundary. This is the single highest-leverage thing you can do in the first few hours — mismatched schemas between AI-1 → AI-2 → AI-3 → Web are the most common cause of "it works alone but not together" on hackathon night.

- **AI-1 → AI-2:** propagated state vectors (position, velocity, epoch, covariance) per object, per timestep, in a fixed frame (agree on ECI as the common frame now).
- **AI-2 → AI-3:** conjunction events with miss distance, relative velocity, combined covariance, object metadata (regime, type, cross-sectional area), and computed Pc.
- **AI-3 → Web:** ranked risk list with Pc, ML risk score, SHAP top-3 features per event, and (for flagged high-risk events) maneuver advisory output.
- **Web → everyone:** one shared JSON schema doc (a single markdown or OpenAPI file in the repo) that all three AI roles write their outputs against. Web owns this file; AI roles propose changes via PR, not Slack messages.

Put this in a shared repo file on Day 1 before any modeling starts.

---

3. Pre-SIH Schedule (Day -5 to Day -1)

Day -5: Setup, Data, and Architecture Lock

- **AI-1:** Set up Celestrak TLE ingestion; get `sgp4` propagating a handful of known satellites; validate against a known pass prediction (e.g. ISS) so you have a correctness baseline from hour one.
- **AI-2:** Read up on Foster's 2D Pc method and the Monte Carlo fallback; sketch the coarse-filter (altitude-band bucketing) + fine-filter (k-d tree distance threshold) approach; no code yet, just the algorithm on paper.
- **AI-3:** Set up LightGBM and TabPFN environments; pull the ESA Kelvins CDM dataset (public, from the 2019 challenge — real conjunction data you can prototype features against before your own pipeline produces any) so you're not blocked waiting on AI-1/AI-2.
- **Web:** Scaffold FastAPI project structure and a bare Cesium.js globe rendering static TLE positions (no live pipeline yet — just prove the render works).
- **Speaker:** Research the problem framing (debris stats, Kessler syndrome, ESA/NASA citations from the blueprint's sources); draft the interface contract doc from Section 2 above with Web.
- **End of day sync (15 min, all 5):** confirm the interface contract is locked. Nothing after today should require changing the data schema.

Day -4: Core Algorithms, First Pass

- **AI-1:** Full TLE ingestion pipeline; propagation working across a demo-scale subset (500–1000 LEO objects, per the blueprint's scale guidance — don't try the full 20,000+ catalog).
- **AI-2:** Implement the coarse+fine conjunction screening filter against AI-1's propagated positions (use AI-1's Day -5 validated output as a fixture even if AI-1 isn't fully done — don't block on integration yet).
- **AI-3:** Feature engineering script against the Kelvins dataset (miss distance, relative velocity, cross-sectional area, orbital regime, object type, historical Pc trend); first LightGBM training run, even on placeholder/public data.
- **Web:** FastAPI endpoints stubbed for each interface contract boundary, returning mock data matching the agreed schema; frontend consumes the mock endpoints (proves the full data path end-to-end before real data exists).
- **Speaker:** Draft the 3-minute demo script structure (problem → live globe → under-the-hood → maneuver advisory → close, per the blueprint); start the pitch deck skeleton; sit in on AI-2's screening implementation for 30 minutes to be able to explain it later.

Day -3: Real Integration Begins

- **AI-1 + AI-2:** First real handoff — AI-2's screening running against AI-1's actual propagated data, not fixtures. Expect bugs here; budget the whole day.

- **AI-3:** Pc calculation from AI-2 becomes an input feature once available; if AI-2 isn't ready by midday, keep training against Kelvins data and swap the data source later — don't let AI-3 sit idle waiting.
- **Web:** Swap mock endpoints for whatever real data AI-1/AI-2 can provide, even partial; get the globe rendering real (not synthetic) satellite positions.
- **Speaker:** QA pass — literally try to break the demo flow by clicking around the dashboard; write down every rough edge for the AI/Web trio to fix. This is more valuable right now than deck polish.

Day -2: ML + Maneuver Layer, Full Pipeline

- **AI-2:** Finish Pc calculation (Foster's method primary, Monte Carlo as fallback/validation check).
- **AI-3:** Retrain LightGBM on real pipeline output (not just Kelvins data) once AI-2's Pc values are flowing; get SHAP plots working; run the TabPFN comparison and document the result (this is your "we benchmarked against a 2026 foundation model" slide — see prior discussion on model choice).
- **AI-3 (second half of day):** Implement maneuver advisory as closed-form delta-v optimization (not a learned model — see below).
- **Web:** Full pipeline connected end to end: ingestion → propagation → conjunction → Pc → ML ranking → maneuver advisory → dashboard, live on the globe.
- **Speaker:** Build the SHAP explainability slide and maneuver advisory slide once AI-3 has real output; rehearse the demo script solo for the first time.

Day -1: Freeze, Rehearse, Buffer

- **All AI + Web:** Morning is bug-fixing only. By midday, freeze the "core engine" as your pre-SIH v1 — no new features, only fixes. This frozen version is your safety net if live-event Wi-Fi or scale problems hit during SIH.
- **Speaker:** Full dry run of the 3-minute pitch with the frozen v1, in front of the whole team; collect feedback; record the backup demo video (per the blueprint's risk mitigation — live 3D rendering on hackathon Wi-Fi is genuinely risky, always have a recorded fallback).
- **All 5:** End-of-day meeting — explicitly list what's "done and frozen" vs "planned for the 24-hour window." Write this list down; it becomes your SIH task board.

---

4. During SIH (24-Hour Window)

With the core engine pre-built and frozen, the 24 hours should go toward the things that genuinely benefit from being done live: adapting to the official problem statement's specific scope or dataset if SIH provides one, live-data hardening, judge-facing polish, and rehearsal — not re-deriving physics you already validated.

Time block	Focus	Primary owners
Hour 0–2	Read the official problem statement carefully; map it against your frozen v1; identify gaps (does the SIH statement	All 5 together

Time block	Focus	Primary owners
	want a specific dataset, region, or constraint your version doesn't handle yet?)	
Hour 2–8	Adapt pipeline to any SIH-specific requirements; AI trio splits by module as before	AI-1/2/3
Hour 2–8	Web hardens dashboard for the adapted scope; adds alert UI polish	Web
Hour 2–8	Speaker rewrites deck/script sections affected by the adaptation; keeps rehearsing the unaffected parts	Speaker
Hour 8–14	Integration pass on the adapted pipeline; this is where bugs concentrate	All, Speaker as QA
Hour 14–18	Feature freeze #2 — stop adding scope, only fix and polish	All
Hour 18–22	Full dry runs of the live demo, back to back; fix whatever breaks	All 5

Time block	Focus	Primary owners
Hour 22–24	Final rehearsal, sleep/food rotation if the format allows it, backup video re-recorded against the final build	Speaker leads, rotate others

Build in real sleep/food rotation if your SIH format allows breaks — a 24-hour event with a team that's non-functional for the last two hours of judging is a worse outcome than a team that rotated rest and stayed sharp for the pitch.

---

5. What NOT to Turn Into a Third ML Model

Worth stating explicitly for the team: the maneuver advisory layer (Section 5.5 of the blueprint) is a constrained optimization problem — given a required increase in miss distance, compute the minimal delta-v burn that achieves it. It's closed-form or a simple numerical search, not something to train a model for. If a judge asks whether that's ML too, the correct answer is "no, that's orbital mechanics" — and being precise about that boundary is part of what makes the rest of the ML claims credible under Q&A.

---

6. Communication Cadence

- 15-minute sync at the start and end of each pre-SIH day (all 5). Keep it to: what shipped, what's blocked, what changed in the interface contract.
- During SIH, shorten to 10-minute syncs every 4 hours — long enough to catch drift, short enough not to eat your build time.
- Any change to the Section 2 interface contract gets flagged to all 5 immediately, not discovered at integration time.